| | | | |
|---|---|---|---|
| **Fiscal Year:** | FY 2026 | **Task Last Updated:** | FY 02/27/2025 |
| **PI Name:** | Stirling, Leia Ph.D. | | |
| **Project Title:** | HCAAM VNSCOR: Responsive Multimodal Human-Automation Communication for Augmenting Human Situation Awareness in Nominal and Off-Nominal Scenarios (80NSSC20K0409) | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H) | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | leias@umich.edu | **Fax:** | FY |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 617-324-7410 |
| **Organization Name:** | University of Michigan | | |
| **PI Address 1:** | Industrial and Operations Engineering | | |
| **PI Address 2:** | 1205 Beal Avenue, G634 IOE Building | | |
| **PI Web Page:** | | | |
| **City:** | Ann Arbor | **State:** | MI |
| **Zip Code:** | 48109 | **Congressional District:** | 12 |
| **Comments:** | NOTE: PI moved to University of Michigan in fall 2019; previous affiliation was Massachusetts Institute of Technology | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2017-2018 HERO 80JSC017N0001-BPBA Topics in Biological, Physiological, and Behavioral Adaptations to Spaceflight. Appendix C |
| **Start Date:** | 12/04/2019 | **End Date:** | 12/03/2024 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 1 |
| **No. of PhD Candidates:** | 4 | **No. of Master' Degrees:** | 2 |
| **No. of Master's Candidates:** | 2 | **No. of Bachelor's Degrees:** | 4 |
| **No. of Bachelor's Candidates:** | 7 | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Whitmire, Alexandra | **Contact Phone:** | |
| **Contact Email:** | alexandra.m.whitmire@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date changed to 12/03/2024 per NSSC information (Ed., 11/10/23). | | |
| **Key Personnel Changes/Previous PI:** | N/A | | |
| **COI Name (Institution):** | Blossom, Jonathon ( NASA Jet Propulsion Laboratory )<br>Atkins, Ella Ph.D. ( Virginia Tech )<br>Liu, Andrew Ph.D. ( Massachusetts Institute of Technology ) | | |
| **Grant/Contract No.:** | 80NSSC20K0409 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

| | |
|---|---|
| **Task Description:** | [Ed. note April 2020: Continuation of "HCAAM VNSCOR: Responsive Multimodal Human-Automation Communication for Augmenting Human Situation Awareness in Nominal and Off-Nominal Scenarios," grant 80NSSC19K0703, with the same Principal Investigator (PI) Leia Stirling, Ph.D., due to PI move to University of Michigan from Massachusetts Institute of Technology in fall 2019]<br>This task is part of the Human Capabilities Assessments for Autonomous Missions (HCAAM) Virtual NASA Specialized Center of Research (VNSCOR).<br><br>Crew extravehicular activity (EVA) is limited on spaceflight missions. Multiple, small robotic spacecraft with varying levels of autonomy are needed to perform tasks that might have been completed by an astronaut (e.g., an exterior surface inspection or repair). Crews on long duration exploration missions (LDEM) will have less access to ground support during task operations. As a result, they will need to process more information and communicate with autonomous robots effectively to ensure tasks are progressing safely and on schedule.<br><br>The objective of these studies is to investigate the use of augmented reality (AR) multimodal interface displays and communication pathways for improving human-robot communication, situation awareness (SA), trust, and task performance. This will lead to developing guidelines for designing human-robot system interactions that enable operational performance for crews on spaceflight missions.<br><br>The specific aims are to:<br><br>1) Develop a simulation testbed for examining communication between human-robot teams.<br><br>2) Develop a hardware testbed for examining communication between human-robot teams.<br><br>3) Evaluate human SA, trust, and task performance within a short duration and long-duration ground-based study (simulation and/or hardware) through testing various interface communication modalities and information displays.<br><br>4) (Option) Perform additional studies for alternate parameters of interest that could be tested using the study testbeds. Additional parameters include timing and persistence of information, gesture command mapping, varying the levels of robot automation, evaluating precision enabled by each command mode. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Augmented Reality (AR) has the opportunity to support decision making across a variety of use-case scenarios, including but not limited to manufacturing, automated vehicles, military training, and entertainment. This research compares AR to other visual modalities for telerobotics applications (considering robotic control, anomaly inspection, and docking) and for astronaut assessment (sensorimotor, dynamic balance, and operational tasks). Results from these studies inform how AR is integrated for task-specific applications, as there may be tasks that have increased benefit from AR, whereas others may have additional considerations that emerge. |
| **Task Progress:** | The objective of these studies was to investigate the use of augmented reality (AR) multimodal interface displays and communication pathways for improving human-robot communication, situation awareness (SA), trust, and task performance. The collection of studies performed supports the development of guidelines for designing human-robot system interactions that enable operational performance for crews on spaceflight missions. Several recommendations can be made based on the studies performed.<br>Our first two studies examined an inspection task in a simulation environment, where a small robotic agent was external to the space station being inspected. Both studies support decreased detection accuracy using AR for this task and increase in detection accuracy during synchronous interactions. Factors influencing the decrease in performance of AR for this task include the reduced contrast and resolution of the presented image, as well as reliability of hand interactions. For tasks that require higher resolution and contrast, like inspection, computer screens are recommend over AR implementations for current technology. For tasks that require short periods of vigilance, synchronous actions between human-robot teams are recommended. However, the optimal range of times for synchronous actions should be further studied.<br><br>We then adapted this simulation environment to examine agent movement as a communication modality. Autonomous agent motion plans can convey system intent to the monitor supporting the decision-making process. In this study, participants viewed an egocentric view from the agent in a supervisory role to determine accuracy and appropriateness of the docking maneuver. Participants appropriately took over control, but the takeover location was dependent on path characteristics. Path characteristics that led to egocentric views with the dock in the primary field of view led to earlier correct takeovers. Larger deviations from a central point-to-point connection were less preferred. Based on these findings, when supervising a robotic agent with an egocentric view, path directionality and heading should support a central view of the target location and paths should minimize deviations from the shortest-line path between waypoints. Path planning methods should incorporate human-aware constraints to support path selection from available solutions.<br><br>In our final set of studies, we further examined how AR impacts sensorimotor and balance related tasks. In these studies, participants performed a task using AR cues versus physical cues. Participants were able to complete all assessment tasks successfully in AR and meaningful measures of performance were obtained from the embedded sensors. These data support that AR has the potential to be used to track performance, BUT the baseline thresholds of performance can differ from other assessment modalities. Based on these studies, we recommend that AR interfaces should: (1) feature wider targets (minimum 0.025 m) than required in physical interfaces, (2) use visual cues to support depth perception (e.g., proximity lighting), and (3) take care in selecting the interaction region for a given object. Improvements are needed in AR to support increased permissibility for hand gesture detection and performance baselines should be determined within AR for use as an assessment platform. |
| **Bibliography Type:** | Description: (Last Updated: 11/25/2025) |
| **Articles in Peer-reviewed Journals** | Weiss H, Stirling L. "Augmented reality assessments to support human spaceflight performance evaluation." Aerosp Med Hum Perform. 2024 Nov;95(11):831-40. https://www.doi.org/10.3357/AMHP.6393.2024 ; PMID: 39711343 , Nov-2024 |

| | |
|---|---|
| **Articles in Peer-reviewed Journals** | Apodaca B, Stirling L, Atkins E. "Orbital dynamic effects on fuel use in sampling-based plans for proximity operations." J Spacecr Rockets. 2025 Mar 16;1-12. https://doi.org/10.2514/1.A35934 , Mar-2025 |
| **Articles in Peer-reviewed Journals** | Larson H, Stirling L. "Formalizing motion plan legibility using empirical manual takeover data in autonomous spacecraft docking." IEEE T Hum-Mach Syst. 2025 Aug;55(4):619-28. https://doi.org/10.1109/THMS.2025.3573243 , Aug-2025 |
| **Articles in Peer-reviewed Journals** | Apodaca B, Helgeson T, Atkins E, Stirling L. "In-orbit space structure inspection trajectory generation." IEEE Robot Autom Lett. 2025 Sep;10(9):8810-7. https://doi.org/10.1109/LRA.2025.3583621 , Sep-2025 |
| **Articles in Peer-reviewed Journals** | Li M, Lee JD. "Purpose outweighs performance: Trust drops more when AI teammates fail to cooperate, but explanations can repair it." Int J Hum Comput Int. 2025 Aug 1;1-22. Online ahead of print. https://doi.org/10.1080/10447318.2025.2539449 , Aug-2025 |